CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)

CSPIT — Department of Computer Engineering · ITUE203 Web Development Frameworks

Student Name	Dhruv Bhoi	**Student ID**	**D26CE154**
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Practical 2: Semantic HTML5 Pages with Accessibility-Ready Structure

Practical Aim & Tools

Aim: Develop static HTML5 skeletons for at least 10 Student_Hub pages such as Home, About, Register, Login, Dashboard, Profile, Events, Directory, FAQ, Contact, Feedback, and Admin using semantic tags with accessibility support.

Tools & Technologies: VS Code, HTML5, WAI-ARIA, Screen Reader Testing

1. Semantic Architecture & WAI-ARIA Landmarks

Modern web standards require semantic HTML5 elements over generic tags to enable screen readers, assistive technology, and search engine spiders to understand page hierarchy. In StudentHub, every page adheres to the following landmark structure:

• <header> & <aside class="app-sidebar">: Application branding, search input, and main navigation landmark.
• <main id="main" class="container">: Primary landmark holding core page content, paired with a <a class="skip-link" href="#main"> for keyboard accessibility.
• <section> & <article>: Grouping logical modules (e.g. event cards, student profile sections, notice boards).
• <footer>: Site metadata, university credentials, and copyright attribution.
• ARIA attributes: aria-live="polite" on dynamic search results and error messages, aria-expanded="false" on accordions and menus, and role="status" on alerts.

2. Minimum 10 Implemented Pages Inventory

StudentHub implements 14 full pages (exceeding the syllabus requirement of 10):

1. index.php (Home / Hero Slider) | 2. about.php (Project & Credentials) | 3. events.php (Catalog & Filters)
4. directory.php (Dependent Filters) | 5. faq.php (Accordion UI) | 6. contact.php (CSV File Storage)
7. feedback.php (Star Rating/JSON) | 8. login.php (Auth & Remember Me) | 9. register.php (Validation & CAPTCHA)
10. dashboard.php (Acade SaaS) | 11. profile.php (User Profile) | 12. students.php (CRUD & CSV Export)

13. manage_events.php (Uploads) | 14. admin/dashboard.php (Charts/RBAC) | 15. db-test.php (Diagnostic Test Suite)

3. Implementation Code: Semantic Shell

Listing: Semantic HTML5 Shell with Skip-Link and WAI-ARIA

```
<!-- Accessible Semantic Structure Skeleton -->
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>StudentHub | Modern Campus Portal</title>
  <link rel="stylesheet" href="assets/css/style.css">
</head>
<body>
  <a class="skip-link" href="#main">Skip to main content</a>
  <div class="app-layout">
    <aside class="app-sidebar" id="appSidebar" aria-label="Main Navigation">
      <a class="sidebar-brand" href="index.php"><div class="sidebar-brand-
icon">▲</div><span>StudentHub</span></a>
      <nav aria-label="Sidebar Menu">
        <ul class="sidebar-nav">
          <li><a href="dashboard.php" class="nav-item active" aria-
current="page"><i>📊</i><span>Dashboard</span></a></li>
          <li><a href="events.php" class="nav-item"><i>📅</i><span>Events</span></a></li>
        </ul>
      </nav>
    </aside>
    <div class="app-main">
      <header class="app-topbar">
        <button class="hamburger" id="hamburger" aria-label="Toggle menu" aria-
expanded="false">☰</button>
        <div class="topbar-search"><input type="search" placeholder="Search events, courses…"
aria-label="Search"></div>
      </header>
      <main id="main" class="container app-content" tabindex="-1">
        <!-- Main page content here -->
      </main>
      <footer class="site-footer"><small>&copy; 2026 StudentHub - Developed by Dhruv Bhoi
(D26CE154)</small></footer>
    </div>
  </div>
</body>
</html>
```

4. Verification Screenshot: Static Home & Navigation

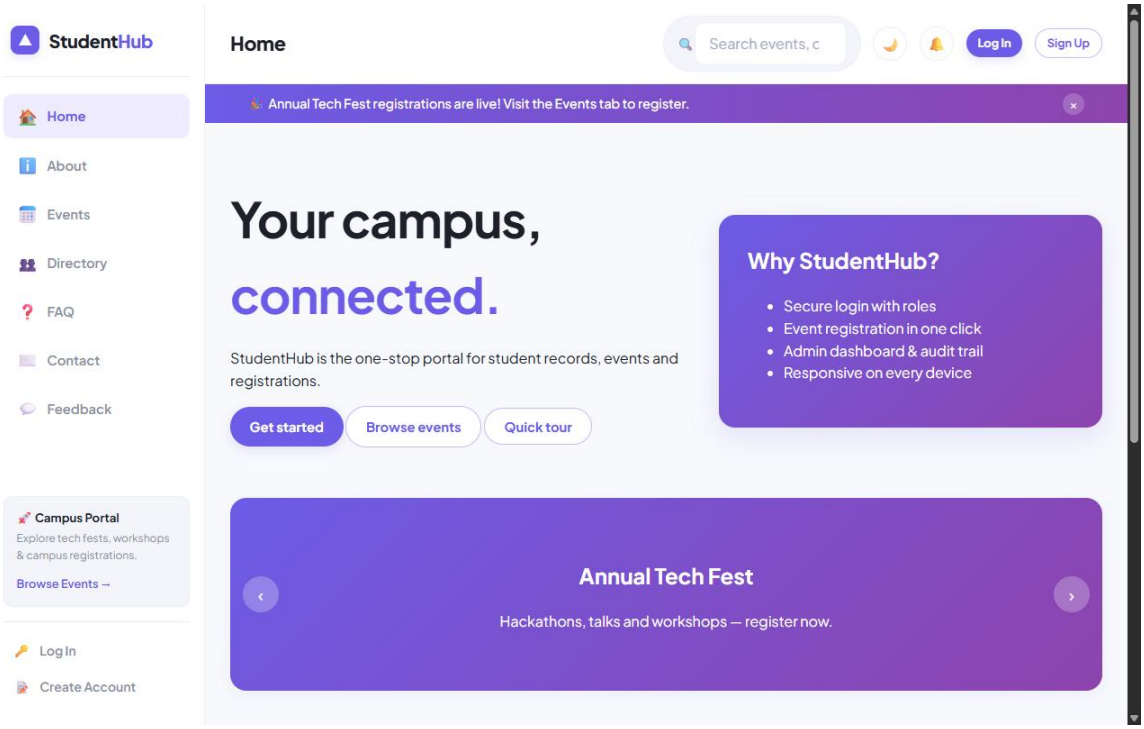


Figure: Semantic Static Landing Page with Accessible Hero and Highlights

5. Key Questions & Analysis

Q: Are semantic elements like header, nav, main, section, article, and footer used properly?

Ans: Yes. StudentHub strictly avoids soup. The page layout uses <header> for titles/tools, <aside> for sidebar navigation, <main> for the single central document body, <article> for self-contained cards (events, reviews), and <footer> for attributions.

Q: Is the website accessibility-ready (WAI-ARIA, skip link, forms)?

Ans: Yes. It implements: 1) Skip link pointing to `#main`, 2) Distinct `<label for="...">` bindings for all form inputs, 3) `aria-live="polite"` regions for live search counts and validation feedback, and 4) `aria-expanded` attributes on interactive menus.

Practical 2 Conclusion & Learning Outcomes

Successfully developed semantic, accessible HTML5 page skeletons for all 14 core modules of StudentHub (exceeding the syllabus requirement of 10). Implemented WAI-ARIA landmarks, keyboard skip links, accessible form labels, and live status regions.